

**Absolute Value and Exponential Functions Performance Task**Form B
Answer key + scoring guidance**DIRECTIONS**

5 items • 8 points • 45 minutes • Form B

Read the scenario, define quantities, show mathematical work, and explain conclusions. The task is scored with an eight-point rubric.

Delivery Operations Model Comparison

Form B. The delivery operations model comparison uses $D(t) = |t - 5| + 1$ for a schedule-deviation score and $V(t) = 900(1.25)^t$ for a dollar-value model, where t is time in years.

1. Identify the vertex, minimum, symmetry, domain, and range of D .

Scoring guidance: Vertex (5,1); minimum 1; axis $t = 5$; domain all modeled t values; range $D \geq 1$.

2. Classify V as growth or decay and state its initial value and percent rate.

Scoring guidance: V shows 25% growth; initial value $V(0) = \$900$.

3. Evaluate both functions at $t = 3$, with correct units.

Scoring guidance: $D(3) = 3$ score units; $V(3) = \$1757.81$.

4. Compare the graphs and rates of change over the modeled domain. Explain why one has a constant multiplicative rate and the other does not.

Scoring guidance: V changes by a constant factor each year. D is piecewise linear with slopes -1 and +1 around its vertex.

5. Defend which model is appropriate for deviation and which is appropriate for value. Address why their output units should not be directly compared.

Scoring guidance: Absolute value models distance from a target; exponential models repeated percent change. The numerical outputs cannot be compared as the same quantity because their units differ.

Eight-Point Rubric

Category	Points	Full-credit evidence
Modeling and setup	2	Defines quantities and creates an appropriate mathematical model.
Accuracy	2	Uses correct procedures and obtains accurate results.
Representation	2	Uses and labels an appropriate graph, table, equation, or structural representation.
Interpretation and justification	2	Explains the conclusion, units, constraints, and reasonableness.